capacity of WDM systems using single-mode fibre (SMF) has reached 100Tbit/s per fibre.

To meet the surging network bandwidth demand, a solution that can be seamlessly integrated with existing optical fibre communication networks is needed. This is where space division multiplexing (SDM) comes in, offering a new dimension in data transmission by enabling multiple cores within a single fibre and utilising multiple wavelength bands through WDM simultaneously.

Space division multiplexing

SDM is a multiplexing technique in fibre-optic communication that uses multiple spatial channels to increase network capacity. These channels operate independently, allowing simultaneous data transmission at the same frequency band within a single fibre. SDM can be categorised into three main types (Fig. 3):

Multicore fibre. Utilises different cores within an optical fibre to transmit data. Its design is more complex and costlier than single-core fibres

Multimode fibre. Uses different modes within an optical fibre, with data received through complex signal processing at the receiving end

Multimode and multicore fibre. A combination of the above techniques to further enhance the data capacity of an optical fibre

SDM in traditional single-core fibres can take various forms:

Reduced coating diameter fibre (RCDF). Maintains the cladding diameter at 125µm while reducing the coating diameter

Reduced cladding fibre (RCF). Reduces the cladding diameter compared to the standard 125/250µm cladding/coating diameter

Few modes fibre (FMF). Utilises three to four individual modes as transmission channels to boost fibre capacity

SDM can also be implemented in single-core fibres by increasing fibre density (i.e., packing multiple fibres within a fibre optic cable) or provisioning multiple modes in a single core for data transmission. However, these methods face capacity constraints, manufacturing complexities, and reduced cost-efficiency, which limit widespread adoption. Consequently, multi-core fibres (MCFs) have been developed to achieve better results with SDM.

Multicore fibre: An overview

MCF refers to an optical fibre containing multiple light-guiding cores within a single strand. It is designed to offer higher bandwidth capacity than